Week 16

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Select the appropriate procedure for a research question among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression. (CLO 8) **2.** Perform a complete hypothesis test, from hypotheses to a conclusion in context. (CLO 8) **3.** Construct and interpret a confidence interval, using simulation or theoretical distributions as appropriate. (CLO 6) **4.** Evaluate whether a study's design allows its conclusion to be generalized to a population. (CLO 1) **5.** Evaluate whether a study's design supports a cause-and-effect conclusion. (CLO 1)	Review Slides: Putting It All Together (with answer key and no-answers versions) All Slides and Interactive Lectures (Weeks 1–15) StatKey	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	Final Exam – Wed Dec 9 (Blackboard, proctored, 2 hours, cumulative; no retake) (all objectives)	Blackboard WileyPLUS StatKey Respondus